

CSE 201 2. Midterm

OlcaY Taner YILDIZ

I. QUESTION (28 POINTS)

Design three classes named Flat, NormalFlat, and DublexFlat to represent flats. The class Flat contains:

- (1 pt) A string data field named **owner** that specify the owner of the flat.
- (1 pt) An integer data field named **roomCount** that specify the number of rooms the flat has.
- (1 pt) An integer data field named **area** that specify the area of the flat.
- (3 pts) Three argument constructor **Flat(String newOwner, int roomCount, int area)**.
- (6 pts) Three getter methods for the three fields **owner, roomCount, area**.

The class NormalFlat extends class flat and contains:

- (1 pt) A boolean data field named **luxury** that specify the luxury of the normal flat.
- (5 pts) Four argument constructor **NormalFlat(String newOwner, int roomCount, int area, boolean luxury)** that calls its parent class's constructor.
- (2 pts) Getter method for the field **luxury**.

The class DublexFlat extends class flat and contains:

- (1 pt) An integer data field named **type** that specify the type of the dublex flat, it can be 0 (basement) or 1 (roof).
- (5 pts) Four argument constructor **DublexFlat(String newOwner, int roomCount, int area, int type)** that calls its parent class's constructor.
- (2 pts) Getter method for the field **type**.

II. QUESTION (50 POINTS)

Design two classes named Building and Street to represent buildings and streets in a city. The class Building contains:

- (1 pt) A string data field named **name** that specify the name of the building.
- (1 pt) An integer data field named **flatCount** that specify the number of flats the building has.
- (1 pt) A boolean data field named **withLift** that specify if the building has a lift.
- (1 pt) An array of flat named **flats** that specify the flats the building has.
- (3 pts) Three argument constructor **Building(String name, int flatCount, boolean withLift)**.
- (4 pts) A method to add a flat to the building **void add(Flat flat)**.
- (5 pts) A method to find the total area of the building **int totalArea()**.
- (8 pts) A method to find the total area of the luxury or roof type flats **int totalAreaRoofLuxury()**.
- (6 pts) Three getter methods for the three fields **name, flats, withLift**.

The class Street contains:

- (1 pt) A string data field named **name** that specify the name of the street.
- (1 pt) An integer data field named **buildingCount** that specify the number of buildings in the street.
- (3 pts) A method to add a building to the street **void add(Building building)**.
- (8 pts) A method to remove a building with a given name from the street **void remove(String buildingName)**.
- (1 pt) An arraylist of building named **buildings** that specify the buildings of the street.
- (2 pts) One argument constructor **Street(String name)**.
- (4 pts) Two getter methods for the two fields **name, buildings**.

III. QUESTION (12 POINTS)

Write a main function using Question 1 to do the following:

- (3 pts) Declare three Flats one owned by Seda, one owned by OlcaY, and the other owned by Sezin where Seda lives in a dublex flat of roof type, Sezin lives in a dublex flat of basement type and OlcaY lives in a luxury normal flat.
- (3 pts) Declare a building named Cse201 with a lift where Seda and OlcaY live.
- (2 pts) Declare a building named Cse202 with no lift where Sezin lives.
- (3 pts) Declare a street named Cse where there are only two buildings namely Cse201 and Cse202.
- (1 pt) Display the total area of building Cse201.

IV. QUESTION (10 POINTS)

Write a static function double totalValue(Street street) which calculates the total value (in TL) of the street with the following constraints:

- The value of a normal flat is calculated as area * 2000 and the value of the dublex flat is calculated as area * 2500.
- The value of a flat increases by 20 percent if it is in a building with a lift.